

# 9th March Experiment Report

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## 1 Closed-Loop Hybrid Control Using Temperature Corrections and Adaptive Optics

### 1.1 Experiment Overview

A hybrid closed-loop control experiment was carried out to stabilise the centroid position of the spectrograph image using a combination of thermal feedback through TEC control and fast corrective action through adaptive optics (AO). The full run lasted for **44.36 hours**, during which centroid position ( $dX$ ,  $dY$ ), optical path length (OPL), TEC setpoint and measured temperature, environmental parameters, and AO actuator commands were recorded continuously.

The experiment consisted of two main regimes:

- **Warmup phase:** Initial stabilisation without feedback, shown as the grey shaded region in the main plot.
- **Hybrid feedback phase:** Combined temperature and AO control used to recover and maintain centroid stability.

The target of the control system was to maintain the centroid position within the sub-pixel regime defined by

$$|drift| = \sqrt{dX^2 + dY^2} \leq 0.10 \text{ px.}$$

Figure 1 summarises the full experiment, showing centroid drift, optical path length evolution, TEC response, environmental conditions, AO cumulative travel, and applied thermal correction steps. During the warmup phase, the centroid moved away from the origin from near-zero values to approximately  $dX \approx -0.33$  px and  $dY \approx -0.54$  px, corresponding to  $|drift| \approx 0.63$  px. After warmup, the centroid drifted further and repeated thermal corrections alone were not sufficient to rapidly restore the centroid(0,0) due to the slower response time of the TEC loop.

The optical path length panels also show that the system experienced noticeable long-timescale variation during the run, while the residual term continued to fluctuate as the feedback attempted to compensate the disturbance. Large excursions in centroid position were therefore corrected first through AO intervention, after which AO and temperature control operated together. In this hybrid mode, AO provided the fast local pull-back of the centroid, whereas TEC control compensated slower thermal drift.

The final state of the experiment, visible at the end of the run, reached

$$dX = -0.0835 \text{ px}, \quad dY = -0.0132 \text{ px},$$

which gives

$$|drift| = 0.0845 \text{ px},$$

placing the final centroid position within the strict 0.10 px target band.

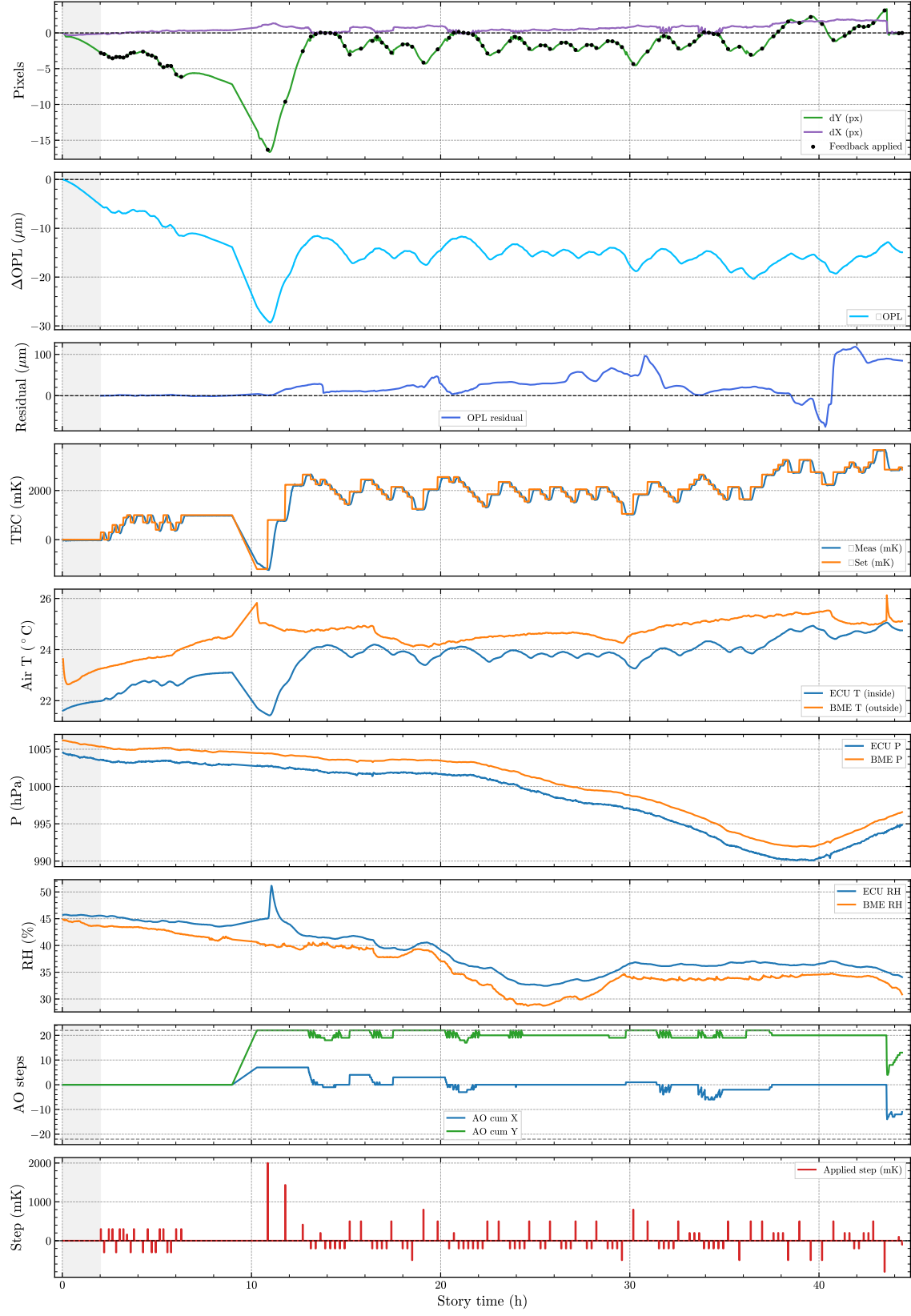


Figure 1: Full hybrid control experiment showing centroid drift, optical path length, TEC response, environmental parameters, AO travel, and applied thermal corrections.

## 1.2 Centroid Distribution in $(dX, dY)$ Space

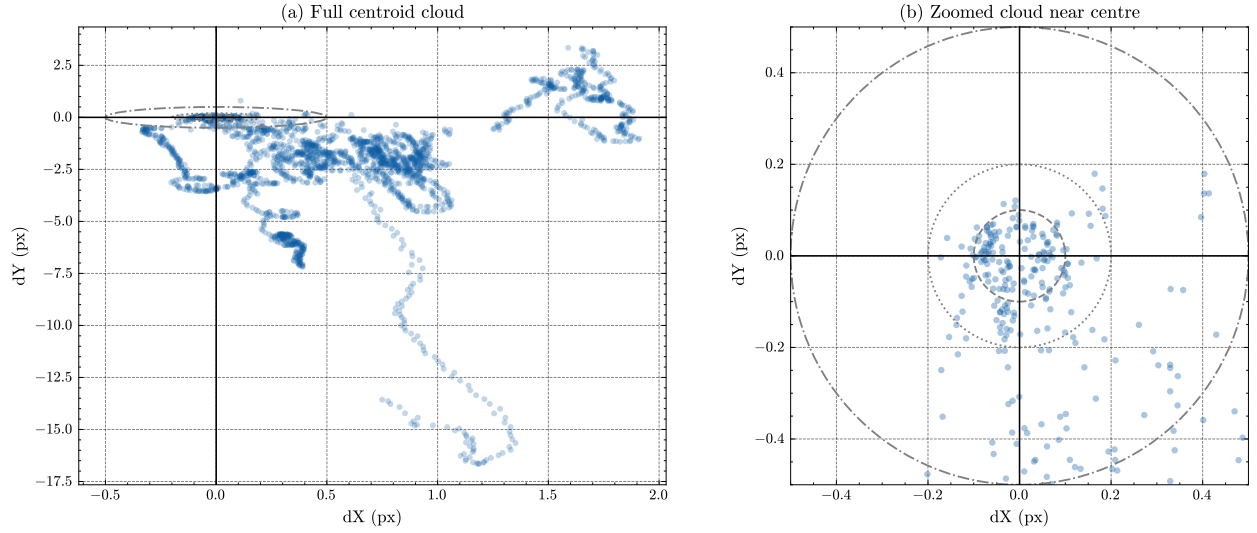


Figure 2: Centroid distribution over the full experiment and near the origin.

Figure 2 shows the distribution of centroid positions in the  $(dX, dY)$  plane. The left panel presents the full excursion over the entire 44.36 h run and shows that the dominant drift occurred mainly along the  $dY$  direction, with additional spread in  $dX$ . The right panel is a zoomed view around the origin, intended to show the cluster of recovered centroid positions after hybrid control was applied. The circles mark the 0.10 px and 0.20 px stability thresholds.

This figure demonstrates two important points. First, the experiment included large drift events and therefore represents a disturbed operating case rather than a quiet laboratory condition. Second, despite these large excursions, the controller was able to bring a subset of points back into the near-origin region, demonstrating recovery to sub-pixel performance after significant centroid displacement.

## 1.3 Quantitative Performance Metrics

The statistical performance of the centroid stability over the full merged run is summarised in Table 1. These values show that the full-run behaviour was dominated by large excursions, particularly in the  $dY$  direction.

| Metric    | Mean Abs (px) | Median Abs (px) | RMS (px) | Std (px) | Max (px) |
|-----------|---------------|-----------------|----------|----------|----------|
| $dX$      | 0.6489        | 0.5166          | 0.8242   | 0.5083   | 1.9111   |
| $dY$      | 2.5565        | 1.8945          | 3.7775   | 2.7815   | 16.6577  |
| $ drift $ | 2.7542        | 2.1305          | 3.8663   | 2.7141   | 16.6998  |

Table 1: Statistical summary of centroid drift over the full merged run.

To quantify fine-control performance more directly, the residence time within selected centroid thresholds was calculated and is listed in Table 2.

| Metric    | Threshold (px) | Fraction of Time | Time (hours) |
|-----------|----------------|------------------|--------------|
| $dX$      | 0.10           | 0.116            | 4.98         |
| $dY$      | 0.10           | 0.074            | 3.16         |
| $ drift $ | 0.10           | 0.051            | 2.20         |
| $dX$      | 0.20           | 0.215            | 9.22         |
| $dY$      | 0.20           | 0.100            | 4.28         |
| $ drift $ | 0.20           | 0.081            | 3.49         |

Table 2: Residence time within selected centroid stability thresholds.

These results show that, over the full 44.36 h disturbed run, the strict radial criterion of  $|drift| \leq 0.10$  px was satisfied for **5.13%** of the time, corresponding to **2.20 hours**. For the individual centroid components, the system remained within  $|dX| \leq 0.10$  px for **4.98 hours** and within  $|dY| \leq 0.10$  px for **3.16 hours**. The relaxed criterion of  $|drift| \leq 0.20$  px was satisfied for **3.49 hours**.

## 1.4 Interpretation Under Environmental Perturbations

The relatively low fraction of time spent within the strict 0.10 px radial stability threshold should be interpreted in the context of the environmental perturbations present during the experiment. The logged outside temperature, pressure, and humidity varied continuously over the run, and these changes are consistent with refractive-index-driven and thermally induced drift within the system.

Previous measurements have shown that the outer enclosure is capable of maintaining temperature stability at the level of approximately **100–200 mK**. Under such improved external isolation, the magnitude of these perturbations would be expected to decrease substantially, and the residence time within the strict sub-pixel band would likely improve correspondingly. The present experiment should therefore be regarded as a conservative disturbed-case test of the controller.

## 1.5 Conclusion

A hybrid closed-loop control strategy combining TEC temperature corrections with adaptive optics actuation was successfully implemented and tested over a 44.36 h experiment. Although the run experienced strong perturbations and large centroid excursions, the controller was able to recover the centroid to the sub-pixel regime by the end of the experiment, achieving a final radial error of

$$|drift| = 0.0845 \text{ px.}$$

The full-run statistics show that maintaining the centroid continuously within the strictest 0.10 px radial threshold remained difficult under the environmental conditions of this experiment. However, the final recovered state, together with the clustering of corrected points near the origin, demonstrates that the hybrid control architecture is capable of restoring sub-pixel stability after substantial drift events. Within the limits of the experimental setup and the external perturbations present during the run, this represents the best achievable performance of the current system configuration.